Ahnaf Shahriar

Email | LinkedIn | Github

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Computer Engineering — *Sept. 2021 – Apr. 2026*

- **GPA**: 3.7
- **Relevant Courses**: Computer Architecture, Distributed Computing, ML for Chip Design, Robot Dynamics & Control, Compilers, Real-Time Operating Systems

Experience

AI HW Engineering Intern — Aug. 2025 – Dec. 2025
Tesla — *Palo Alto, CA*

- **Emulation bringup**: Led GPU bringup in *ZeBu* emulation with *Zephyr ZTest* for firmware integration testing
- **Rust Development**: Built and Linux Rust drivers for Power Supply automations and monitoring for 20+ PSU units
- **Post-silicon validation**: Designed automated validation scripts that exercises FLIT mode *PCIe* TLP and LTSSM testing
- **Performance & power characterization**: Profiled end-to-end Camera workloads and implemented tests to characterize performance and power during early bringup.

Wireless Firmware Developer — Jan. 2025 – Apr. 2025
ON Semiconductor Corporation — *Waterloo, ON*

- **SDK Development**: Integrated *I3C* drivers and *Bluetooth Low Energy* profile services into sample code.
- **Wireless Connectivity**: Developed and integrated cutting-edge *LE Audio, HFP, and A2DP* functionality into an all-in-one Application.
- **Wireless Manager** Designed a modular state machine for bluetooth, improving power efficiency by *25%*
- **Test Automation**: Automated SDK release test cases, reducing testing time by up to *80%* for samples

Embedded Software Engineering Intern — Sept. 2022 – Dec. 2022
Synapse Product Development — *Seattle, WA*

- **Prototyping**: Leveraged Zephyr RTOS to create a proof of concept on *NRF52 BLE* device.
- **Python APIs**: Designd custom lab automation APIs for equipment from *Agilent, Keysight, NI, Tektronik.*
- **Automation**: Streamlined testing and in house procedures using *Python* and *Bash.*
- **Driver Development**: Wrote low-level *I2C and UART* drivers for in-house PCB test platforms on Zephyr RTOS.

Firmware Developer — Jan. 2022 – April 2022
Ford Motor Company of Canada — *Remote*

- **Unity/Cmock Test framework**: Lead developer for optimization for unit testing, achieving up to *30% faster* runtime while using *50%* less manually written test cases.
- **Automation**: Improved *Jenkins CI/CD* pipelines to support unit testing automation using *Python*
- **Embedded Trace Debugging**: Built high speed Parsing tools to parse through CAN and Serial messages exceeding *10 million+* lines.

Projects

LC-3 Emulator: A C emulator for an educational ISA. Improves by 25% on research paper using *Python* data logging.

Real Time Executable: A RTOS implementation in STM32 capable of Pre-emptive task switching and its own Malloc

Autonomous Seeker: An Autonomus RC vehicle equipped with a pick and place robot. Created to Compete in Quidditch!

VHDL Compiler: A Java Compiler for creating VHDL circuits. Using a boolean intermediate representation

Technical Skills

Languages: C/C++, Java, Python, Tcl, Bash scripting, ASM, VHDL, SystemVerilog/Verilog
Tools: OpenCV, ROS, Linux, Qemu, LLDB/GDB, AWS, Docker, WireShark, UVM, Matlab
Hardware: Microcontrollers (ESP32, STM32, nRF, Arduino, PIC), Oscilloscopes, Logic Analyzer, Spectrum Analyzer
Protocols: TCP/IP, Ethernet, CAN/CAN-FD, LIN, AXI, Serial(I2C, SPI, I3C, UART), BLE, LoRan